

Research Statement

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Uncertainty-Aware Scientific Machine Learning for Engineering Systems

Modern engineering increasingly depends on computational models that must remain trustworthy when physical conditions vary, measurements are noisy or incomplete, and high-fidelity simulation is too expensive for repeated decision making. My research asks a central question: how can we build models that infer hidden physical quantities, discover missing governing structure, and make fast predictions while representing uncertainty honestly? I address this question by combining Bayesian inverse modeling, stochastic physics discovery, operator learning, and engineering simulation. Advanced manufacturing is my primary proving ground because it concentrates the challenges that motivate my methods, including coupled physics, heterogeneous sensing, expensive experiments, process variability, and quality-critical decisions. The underlying methods, however, are designed to transfer to materials, semiconductor, thermal, aerospace, energy, and other engineered systems. My long-term goal is to establish a research program in uncertainty-aware scientific machine learning that connects four capabilities: infer latent states and parameters, discover governing structure, predict system response rapidly, and allocate sensing or design decisions under uncertainty.

Foundation: inference, discovery, and fast physics-based prediction

Bayesian inverse modeling for hidden physical quantities. Inverse problems ask what physical state, parameter, or mechanism produced the measurements we can observe. They are often ill-posed, and the difficulty grows when observations vary across space and time or when measurement and model errors are conflated. My work on the inverse heat-conduction problem developed a Bayesian spatiotemporal modeling approach that integrates heat-transfer structure with statistical inference to recover unobserved thermal quantities from sparse measurements. Published in the ASME Journal of Heat and Mass Transfer, this work established a principle that continues to guide my research: physically meaningful inverse solutions should represent uncertainty rather than return a single overconfident estimate. I have extended this principle to instrumented indentation, where stochastic inverse modeling is used to recover elastoplastic properties from load-displacement information. The current extension of this work, developed with collaborators at NIST and Oak Ridge National Laboratory, focuses on data-consistent characterization in the presence of process-dependent material variability and imperfect surrogate predictions.

Physics discovery in the presence of uncertainty. Accurate parameter estimation is insufficient when the assumed governing model is itself incomplete. To address model-form uncertainty, I developed the stochastic inverse physics-discovery framework reported in Communications Physics in 2026. The method treats unknown coefficients in candidate governing equations as random variables and infers distributions whose push-forward matches the empirical distribution of observed system responses. Across ten scenarios spanning nonlinear oscillators, chaotic systems, ecological data, and infiltration dynamics, the framework reduced coefficient error by about 80% relative to seven comparison methods and achieved exact structural identification in seven of ten cases. This contribution reframes governing-equation discovery as an uncertainty-aware inverse problem rather than a noise-sensitive curve-fitting exercise. A complementary direction in my work investigates how counterfactual explanations can be used to interrogate nonlinear dynamical models and expose the structure responsible for observed behavior.

Operator learning for scalable simulation and design. Many engineering workflows remain too slow for repeated inference, optimization, or control because every candidate decision requires another high-

fidelity simulation. My CRONet work addresses this bottleneck using an operator-learning surrogate for finite-element analysis within topology optimization. Published in *Manufacturing Letters*, CRONet learns the evolution of design fields and demonstrates that learned operators can reduce the computational burden of iterative physics-based design while preserving close agreement with finite-element benchmark solutions. This work provides the third foundation of my agenda: trustworthy engineering AI requires not only principled uncertainty treatment, but also fast surrogate models that preserve the structure of the underlying physical problem.

Research thrust 1: Data-consistent inference from multimodal sensing

My first faculty thrust will develop probabilistic models that update from heterogeneous observations while separating changes in the physical system from sensor bias, measurement noise, intrinsic process variability, numerical error, and model discrepancy. The methodological core will be hierarchical Bayesian and data-consistent inversion in which latent physical parameters, sensor-specific error processes, and discrepancy terms are inferred jointly. Gaussian-process and neural-operator surrogates will make repeated inference computationally feasible, while sequential inference will support streaming data. Initial demonstrators will emphasize manufacturing and materials processes where thermal, force, acoustic, image, and geometric measurements provide partial and sometimes conflicting views of the system state.

The scientific objective is calibrated decision support rather than prediction alone. A useful model should report whether a parameter is identifiable, whether uncertainty is dominated by sensing or physical variability, and whether additional data would materially change a downstream decision. I will therefore connect posterior uncertainty to experimental design and value-of-information criteria so that sensing effort is directed toward the measurements that are most consequential. This creates a natural bridge between statistical inference and operations-research ideas in sequential decision making. The same framework can support digital-twin applications in which model calibration is not a one-time offline activity but a continuous process of assimilation, diagnosis, and updating.

Research thrust 2: Self-correcting models through physics discovery and active experimentation

My second thrust addresses the question of what a model should do when new observations cannot be explained by its assumed physics. I will integrate stochastic physics discovery with change detection and active experimentation to create self-correcting engineering models. When observed responses are statistically inconsistent with the predicted distribution, the framework will first test whether the discrepancy can be explained by parameter drift, sensing error, or known variability. If not, a physics-discovery layer will search for missing or altered governing terms. Competing physical explanations will then be evaluated through uncertainty-aware experimental design that selects the next sensing location, loading condition, or process setting with the highest ability to discriminate among them.

Autonomous material characterization will provide an initial testbed. I will integrate stochastic inverse modeling with an indentation workflow in which an active-learning policy selects informative experimental or simulation conditions, inverse modeling estimates elastoplastic parameters, and the discovery layer evaluates whether the assumed constitutive model remains adequate. The broader contribution will be a general architecture in which sensing, parameter identification, physics discovery, and experiment selection form a closed loop. This direction is especially important for autonomous and cyber-physical systems because reliable autonomy requires models that can recognize, explain, and respond to deviations from the assumptions under which they were trained.

Research thrust 3: Neural operators for rapid design, optimization, and engineering decisions

My third thrust will extend operator learning from topology optimization toward uncertainty-aware surrogates for coupled engineering systems. The first stage will address higher-resolution and three-dimensional problems, temperature-dependent properties, and coupled thermo-mechanical constraints. I will then develop operator models that map uncertain boundary conditions, manufacturing variability, and evolving system state to distributions of performance rather than single deterministic predictions. Embedding these surrogates within inverse problems, optimization, and receding-horizon decision workflows will enable rapid design updates without repeatedly solving the full governing equations.

A critical research question is when a learned operator should be trusted outside its training distribution. I will combine calibration diagnostics, sensitivity analysis, active sampling, and multi-fidelity correction to identify regions where surrogate predictions can be accepted, corrected with high-fidelity simulation, or rejected. The intended outcome is a hierarchy in which fast learned models work with, rather than replace, governing physics. This provides a pathway from component-level surrogates to system-level digital representations whose computational fidelity adapts to the decision being made.

Integrated program, partnerships, and research culture

Together, these thrusts form an infer-discover-predict-decide loop: Bayesian inversion estimates what cannot be measured directly; physics discovery identifies when the assumed model is incomplete; operator learning accelerates prediction; and active experimentation or optimization allocates resources where they have the highest decision value. My five-year objective is to develop this loop into an open, modular scientific-machine-learning platform validated from analytical benchmarks through controlled experiments and partner-driven engineering problems. The program is intentionally interdisciplinary and creates natural collaborations with researchers in statistics, operations research, dynamical systems, controls, manufacturing, materials, computing, and systems engineering. Existing interactions with NIST and Oak Ridge National Laboratory provide a foundation for national-laboratory collaboration, while manufacturing, semiconductor, aerospace, and energy applications create opportunities for industry partnerships and externally funded research. I intend to pursue early-career opportunities at NSF and DOE and to build larger collaborative proposals as the experimental and computational platform matures.

The research group I build will treat reproducibility and mentorship as technical infrastructure. Software, documented benchmarks, version-controlled computational workflows, and FAIR data practices will make results auditable and lower the barrier for new students to contribute. I will use a tiered mentoring structure in which doctoral researchers mentor master's and undergraduate students while receiving explicit training in supervision and research communication. Students will begin with tractable modules, present results early, and progressively assume ownership of methodological questions. Remote and hybrid projects will provide additional entry points for students who have limited access to research-intensive environments. My goal is to train researchers who can move fluently among physical reasoning, statistical modeling, computation, and engineering decision making.

The broader vision is a class of engineering models that remain useful not because they are assumed to be correct, but because they can quantify what they do not know, identify when their assumptions fail, learn what physics is missing, and update decisions accordingly. By advancing the statistical and computational foundations of this capability, I aim to contribute methods that are both rigorous enough for scientific discovery and practical enough for next-generation engineering systems.